

A Hybrid Deep Learning Model for Retinal Disease Classification

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**A Capstone project report submitted in partial fulfillment of
the requirements for the degree of Bachelor of Science in
Information and Communication Engineering**



**Department of Computer Science and Engineering
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Declaration

We, **Fahim Rahman, Mahian Nawaj, Nayeem Mahmud Sakib and Farhan Ishraq** hereby, declare that the work presented in this capstone project report is the outcome of the investigation performed by us under the supervision of **Dr. Anup Kumar Paul** , Associate Professor, Department of Computer Science and engineering, East West University. We also declare that no part of this project has been or is being submitted elsewhere for the award of any degree or diploma, except for publication.

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Letter of Acceptance

The capstone project report entitled "**A Hybrid Deep Learning Model for Retinal Disease Classification** " is submitted by **Fahim Rahman, Mahian Nawaj, Nayeem Mahmud Sakib** to the Department of Computer Science and Engineering, East West University, Dhaka, Bangladesh is accepted for the partial fulfillment of the requirement for the degree of Bachelor of Science in Information and Communication Engineering on 22th February 2025

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Abstract

The accurate and early detection of retinal diseases is critical for effective treatment and prevention of vision loss. Optical coherence tomography (OCT) imaging has become an essential tool for non-invasive diagnosis of retinal diseases. Clinicians now have access to more data than they can effectively parse and digest. Investigating state-of-art diagnostic techniques, This thesis addresses the problem of improving both retinal layer segmentation and classification accuracy and efficiency by exploring multiple convolutional Neural Network(CNN) and Transformer based architectures. This study focuses on a four-class classification problem to automatically detect Choroidal Neovascularization(CNV), Diabetic Macular Edema(DME), DRUSEN and NORMAL class in OCT images. The primary research question is: Which CNN architecture or combination of architectures provides the most effective solution for accurately classify the retinal diseases in OCT images across different retinal diseases? To answer this, we implemented a feature fusion model, which combines complementary strengths of two state-of-the-art architectures. Unet++ and Swin transformer. A public dataset of 84,495 gray scale OCT images was used to evaluated through empirical testing based on key metrics such as accuracy, precision, recall and f1-score. The results demonstrate that the model achieved an outstanding 97% accuracy, outperforming individual CNN models in both accuracy and robustness. This thesis contributes to both the theoretical understanding of feature fusion in CNNs and its practical applications in real-time systems. Future work could focus on optimizing computational efficiency and testing the model across more diverse datasets. The originality of this research lies in the effective use of feature fusion to enhance diagnose complicated multi-disease in an eye.

Acknowledgments

This project marks a significant milestone in our academic journey, and we would like to extend our heartfelt gratitude to all who contributed to its success. The work was carried out at the Department of Computer Science and Engineering, East West University, Bangladesh.

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We are grateful to East West University for providing the necessary lab facilities, allowing us to train our models and conduct experiments. The collaborative spirit of our capstone team—Mahian Nawaj, Farhan Ishraq, Nayeem Mahmud Sakib and my self Fahim Rahman—was key to overcoming challenges and completing this project successfully.

Lastly, we would like to thank our families for their constant support and encouragement, without which this accomplishment would not have been possible.

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List of Algorithms

- **MobileNetV3**
A lightweight and efficient deep learning model optimized for mobile and edge devices.
- **Vit-tiny**
A compact vision transformer model that applies self-attention for image classification with fewer parameters.
- **DenseNet21**
A densely connected convolutional network that enhance feature reuse for efficient deep learning.
- **InceptionResNet**
A hybrid architecture combining Inception modules with residual connections for improved accuracy and efficiency.
- **Cross-Entropy Loss Function**
A loss function commonly used in classification problems, measuring the difference between the true distribution (actual labels) and the predicted distribution (model outputs).
- **Hybrid Unet-Swin Algorithm**
A method that combines features extracted from multiple models to enhance performance, particularly in OCT image tasks. This algorithm leverages the strengths of different architectures to create a more comprehensive feature set.

List of Acronyms

- **CNN:** Convolutional Neural Network
A class of deep neural networks primarily used for analysing visual imagery.
- **OCT:** Optical Coherence Tomography
Refers to non-invasive imaging technique that captures high-resolution cross-sectional images of biological tissues.
- **DL:** Deep Learning
A subset of artificial intelligence that enables systems to learn from data and improve from experience.
- **F1 Score:** A measure of a test's accuracy that considers both precision and recall to compute the score.

Chapter 1

Introduction

1.1 Background

Automated Optical Coherence Tomography (OCT) Image Analysis using Convolutional Neural Networks (CNNs) presents a promising approach to enhancing the early detection and diagnosis of retinal diseases. This project focuses on developing a deep learning model capable of classifying OCT images into distinct categories such as Normal, Choroidal Neovascularization (CNV), Diabetic Macular Edema (DME), and Drusen. The proposed model employs a combination of a segmentation unit and a pre-trained CNN architecture, fine-tuned to achieve high accuracy and reliability in retinal disease classification. The integration of advanced image preprocessing techniques and data augmentation ensures robust model performance.

The accurate and early detection of retinal diseases is critical for effective treatment and prevention of vision loss. Optical coherence tomography (OCT) imaging has become an essential tool for non-invasive diagnosis of retinal diseases. The importance of segmentation of retinal layers in OCT is that it allows the diagnosis and evaluation of various retinal and systemic diseases.

Today, the field of ophthalmology is experiencing rapid progress in the development of medical diagnostic tools. However, decision-making in this area requires the analysis of large amounts of information, which can be a complex and time-consuming task for doctors and other professionals. One of the important tools for diagnosing eye diseases is the analysis of OCT images

1.2 Problem Statement and Analysis

Retinal pathologies are given intensive and widespread attention with a common understanding that such a problem is among the leading causes of severe vision loss and blindness on the global level. Traditionally, the clinician examines every cross-section within OCT individually, which rises the workload of the Optometrists and heavily causes misdiagnosis and border the capacity of producing a large number of ground truth. Developing automated diagnosis methods for retinal diseases are providing enormous benefit.

This project addresses the question: "How can we leverage feature fusion from multiple deep learning models to enhance the accuracy and robustness of automated retinal disease classification without significantly increasing computational overhead?" By conducting a comparative analysis of existing models and proposing a new feature fusion strategy, we aim to solve the issues of model limitations and enhance real-time application capabilities.

1.3 Project Objectives

The primary objectives of this capstone project are as follows:

- To design and implement a disease classification system using the **hybrid** model.
- To evaluate the performance of the system in terms of accuracy, precision, recall, and F1-score across a variety of hand gesture datasets.
- To optimize computational efficiency, reducing training time and memory usage while maintaining or improving performance.
- To provide a comprehensive comparison of the proposed model with other state-of-the-art OCT models.

1.4 Project Contributions

This project contributes significantly to the field of Ophthalmology and computer vision interaction. The key contributions are:

1. A novel hybrid model, that combines Unet++ and swin transformer, leading to significant improvements in classification accuracy.
2. A thorough performance comparison between **Our model** and existing CNN architectures, showcasing the advantages of multi-model feature fusion.
3. Practical applications of the proposed system in areas such as retinopathy.

1.5 Project Outlines

This thesis is structured as follows:

- **Chapter 2** provides a literature review on existing OCT techniques, focusing on various CNN architectures and other strategies.
- **Chapter 3** discusses the methodology employed in this project, detailing the dataset, preprocessing techniques, model architecture, and training processes.
- **Chapter 4** presents a comprehensive analysis of the experimental results and their implications. This chapter includes performance metrics, comparisons between the proposed hybrid model and other CNN architectures, as well as a discussion on the significance of the findings in real-world applications.
- **Chapter 5** concludes the thesis by summarizing the key outcomes and suggesting directions for future research.

Chapter 2

Related Works

2.1 Retinal Disease Detection Using Transfer Learning Techniques

The rationale behind choosing transfer learning stems from the limitations of conventional deep learning methods employed by other researchers. These methods often demand a substantial amount of OCT data for effective training of models. To surmount this challenge, some researchers adopted a transfer learning approaches. By utilizing transfer learning, efficiency can be optimized, data requirements reduced, and the model training process expedited, leading to significantly improved performance outcomes and more robust diagnostic tools.

2.2 Deep learning Models for Retinal Disease Detection

A range of studies is proposed for eye diseases diagnosis focused on deep learning. Suggested the use of transfer learning using VGG16, ConvNets and inception V3 models to identify the four eye, retinal disease groups, by analysing the OCT image.

- **DenseNet169** (Elena-Anca., 2023) introduced dense connections between layers, promoting feature reuse and mitigating the vanishing gradient problem. However, its computational complexity limits its efficiency in real-time applications.
- **OpticNet** (Pavitra K C, Preetam Kumar and Geeha M, 2022) uses “Statistical Analysis of deep learning models for classification using OCT images”. The model incorporated a categorical-cross entropy loss function. The optimizer need to reduce the loss function,
- **CTS-Net** (Andrey Pershin, Vailii Borisov, 2023) In this work authors built a CTS-Net model based on the underlying CSWin transformer skeleton to implement pixel-level segmentation.
- **ResNetV2**(Xueshen Li, Xinlong Hou) The study relies on the ORIGA dataset which may limit its generalizability.
- **Inception V3** (Aya Adel, Mona M.Soliman) uses depthwise separable convolutions to reduce computational complexity, although the model’s performance could be further validated on a larger dataset.

2.3 Identified Gaps and Proposed Approach

Despite the progress in CNN architectures, several gaps remain:

- **Lack of Temporal Dynamics:** Many models, such as VGG19 and DenseNet169, excel at spatial feature extraction but struggle with temporal features crucial for cross section images of retina.
- **Real-Time Efficiency:** Models like ResNet101V2, while powerful, face challenges in real-time applications due to their high computational complexity.
- **Generalization Across Diverse Environments:** Several models struggle to generalize across diverse dataset and low resolution images.

To address these challenges, we propose a hybrid strategy combining U-net++ and Swin transformer. This approach leverages the highlighting the region of interest with Unet++ incorporating Shifted window based architecture Swin for computational efficiency. The goal is to enhance real-time accuracy and robustness in varied environments.

2.4 Applications and Challenges

Retinal disease classification using deep learning model has revolutionized ophthalmology by enabling faster, more accurate, and automated diagnosis of various retinal conditions. The applications could be early detection and diagnosis of Retinal Diseases. Automated screening in Rural and Underserved Areas. Assisting Ophthalmologists and reducing workload. Personalized Treatment Planning and Disease Progression Monitoring as well as Personalized Treatment planning and disease progression monitoring. There are few challenges like Different patients have varying eye structures, lighting conditions, and imaging artifacts. Some diseases have fewer samples leading to biased model predictions.

2.5 Recent Advances

Recent advancements in retinal disease classification have significantly enhanced early detection and treatment strategies. These developments encompass novel deep learning architectures, integration of advanced imaging modalities and the application of artificial intelligence in clinical settings.

Chapter 3

Materials and Method

3.1 Materials

3.1.1 Dataset Collection

3.1.2 The dataset we used in this project is Kermany OCT2017 dataset, which is commonly used for medical imaging tasks, particularly in ophthalmology. This dataset focuses on OCT scans and is designed for tasks such as image classification and segmentation. The dataset originates from the research by Daniel Shu Wei Ting, titled “Development and Validation of a Deep Learning System for Diabetic Retinopathy and Related Eye Diseases Using Retinal Images from Multiethnic Populations With Diabetes”. The dataset is organized into folders by classes, making it suitable for use with PyTorch’s ImageFolder. The image categories are NORMAL (Healthy scans with no abnormalities), CNV (Choroidal Neovascularization), DME (Diabetic Macular Edema) and DRUSEN. 4,500 test sample.

3.1.3 Dataset Exploration

The dataset was explored using visualization techniques to understand its composition and distribution. Each image in the dataset was carefully examined to ensure proper labelling and balanced representation.

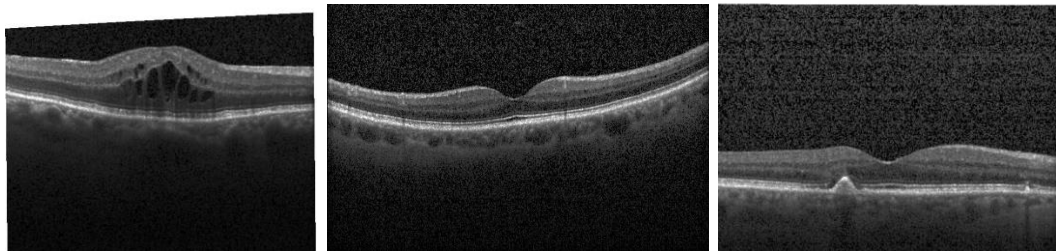


Figure 1: Sample images from the dataset representing the cross section of retina.

3.1.4 Dataset Sampling

The dataset is organized into three main directories, Train, Validation(val), test. Each directory contains subfolders for the four categories. The images are grayscale and often

varying resolutions requiring resizing. In total the Dataset contain 84,495 images. 82,000 training samples, 1200 validation sample and

3.1.5 Dataset Processing

The data preprocessing steps involve transforming the images to prepare them for input to the model. These steps are essential to ensure that the data is in the correct format and to improve the performance of our model. Here's a breakdown of the preprocessing steps:

Image Transformation: the transformations are applied to each image in the dataset to ensure consistent input shape, normalization and appropriate data augmentation.

Image Resize: `transforms.Resize((224,224))` resize all images to the target dimensions of 224*224 pixels. This is a common size for most pre-trained models and is suitable for deep learning models to maintain a consistent input size.

Transforms to Tensor: This step converts the image from a NumPy array into a PyTorch tensor. This step is required because Pytorch models expect inputs to be in tensor format, and it also ensures the data is properly aligned in memory for optimization.

Transforms to Normalize: This step normalizes the image data to have a mean of 0 and a standard deviation of 1 for each channel(Red, Green, and Blue). Deep learning models perform better when the input features are normalized, the model can learn more effectively.

Create Datasets ImageFolders: This setup automatically labels the images based on their folder names. Each folder in training, validation and test directories corresponds to a class and the images inside those folders are labeled with the folder name as their target class.

Splitting the training Dataset: The step is used to split the training data into training and validation sets, ensuring validation dataset have enough image for model evaluation during training.

Batch and Shuffle: Loading data in batched is essential for efficient training as it reduces the memory footprint and speeds up training. And Shuffling the training data helps the model generalize better by avoiding bias from the order of the data.

3.1.6 Research Environment and Devices

The project was executed in the **East West University** Big Data Analytics lab under supervision, utilizing **NVIDIA GPU**-enabled machines for accelerated model training and for few testing the Kaggle **Accelerator NVIDIA GPU kernel-Tesla P100 version 1** are used. The models were developed using the **PyTorch** libraries.

3.2 Method

3.2.1 Proposed Model

The feature fusion model, combines **Unet++** and **Swin Transformer** models to enhance OCT image classification accuracy. By integrating features extracted from the penultimate layers of these models, we achieve high classification accuracy.

Segmentation Model: Unet++

The purpose of this model is to extract segmentation maps that highlight the region of interest. The backbone of this architecture has ResNet-34 encoder pretrained on ImageNet. The model implies skip connections to retain spatial information at multiple scales also deep supervision for better segmentation results. Some modifications are performed like output is a single-channel binary map representing the probability of the region belonging to a diseased area. The segmentation map is passed through a sigmoid activation function also the output resized using bilinear interpolation to match the input size (224*224) for integration with the classification model.

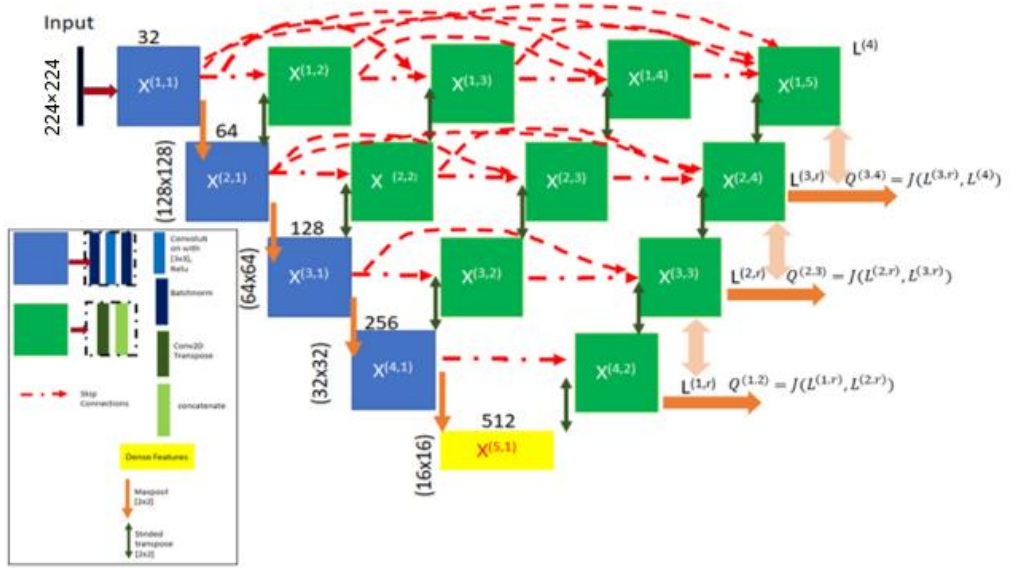


Figure 2: Architecture of U-net++

Classification Model: Swin Transformer The model classifies the input image into one of the four categories using both global and localized features. The transformer works on few key components like Patch Partition, Linear Embedding, Transformer Blocks, Bottleneck Layer and a Classification Head

Patch Partition: The Patch partition process in the Swin Transformer is a critical step for converting an input image into a format suitable for processing by the model. The input to the model is an image of size $H \times W \times CH$. Where H, W and C are the Height, Width and Number of color channels(3 for RGB images). The image is divided into non-overlapping patches of size $P \times P$, where P is the patch size. Here it is $P=4$.

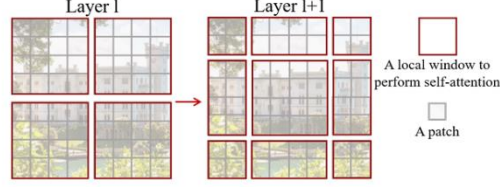


Fig3: An illustration of shifted window approach

In layer l (left), a regular window partitioning scheme is adopted, and self-attention is computed within each window. In the next layer $l + 1$ (right), the window partitioning is shifted, resulting in new windows. The self-attention computation in the new windows crosses the boundaries of the previous windows in layer l , providing connections among them. In our custom architecture the image splits into non overlapping patches using a 2D convolution with a kernel size of 4×4 times 4×4 , stride 4, and padding 0. In output each patch is represented by a 64-dimensional feature vector. We added a batchNorm2D layer for normalizing the output and a ReLU activation to introduce non-linearity. In a typical Swin Transformer, patch embedding is performed with a 4×4 window. The simplification here avoids more advanced overlapping patch embeddings for computational efficiency. The pretrained Swin transformer uses PatchEmbedding and LayerNorm to encode image patches and normalizes them using layer normalization. Here, BatchNorm2d is used because it is more efficient on GPUs for convolution-based architectures. For resource-constrained setups, this setup reduces memory and computational overhead while retaining the main functionality of patch extraction.

Swin Transformer block: Swin transformer is built by replacing the standard multi-head self attention(MSA) module in a transformer block by a module based on shifted windows. With other layers kept the same. As illustrated bellow-

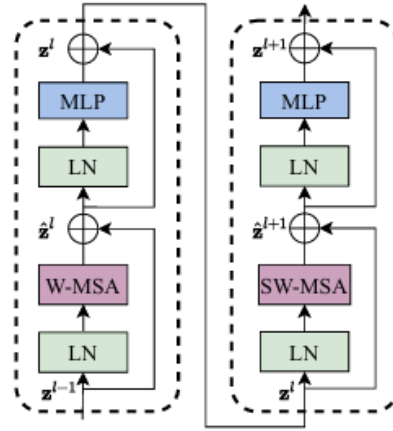


Fig4: Two Successive Swin Transformer Blocks

The swin transformer block consists of a shifted window based MSA module followed by a 2-layer MLP with GELU non-linearity in between. A layerNorm(LN) layer is applied before each MSA module and each MLP, and a residual connection is applied after each module.

Self-attention in non-overlapped widows: To compute self-attention within local windows. The windows are arranged to evenly partition the image in a non-overlapping manner. Supposing each window contains $M \times M$ patches, the computational complexity of a global MSA module and a window based one on an image of $h \times w$ patches are:

$$(MSA) = 4hwC^2 + 2(hw)^2C, \quad (1)$$

$$(W-MSA) = 4hwC^2 + 2M^2hwC, \quad (2)$$

Global self-attention computation is generally unaffordable for a large hw , while window based self-attention is scalable.

Shifted Window Partitioning in successive blocks: The window-based self-attention module lacks connections across windows, which limit its modeling power. A shifted window partitioning approach which alternates between two partitioning configurations in consecutive Swin Transformer blocks. Then the next module adopts a windowing configuration that is shifted from that of the preceding layer, by displacing the windows.

Swin Architecture: An overview of the Swin Transformer architecture is presented in Figure bellow, It first Splits an input image into non-overlapping patches by a patch splitting module, like ViT. Each patch is treated as a “token” and its feature is set as a concatenatin of the raw pixel RGB values. In our implementation, we use a patch size of 4×4 and thus the feature dimension of each patch is $4 \times 4 \times 3 = 48$. A linear embedding layer is applied on this raw-valued feature to project it to an arbitrary dimension denoted as C .

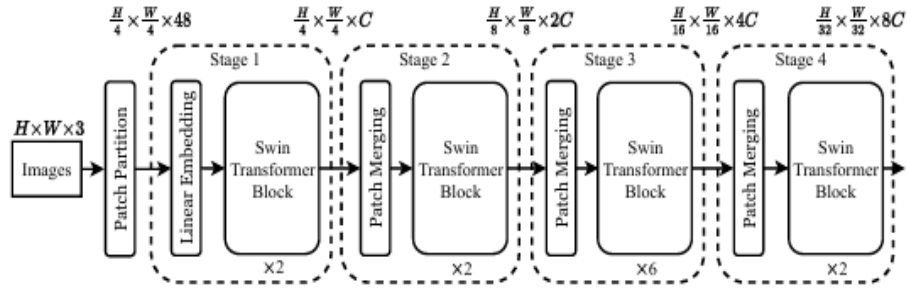


Fig5: The architecture of a Swin Transformer

Several Transformer blocks with modified self-attention computation are applied on these patch tokens. The Transformer blocks maintain the number of tokens($H/4W/4$), and together with the linear embedding are referred to “Stage 1”.

To Produce a hierarchical representation, the number of tokens is reduced by patch margin layers as the network gets deeper. The first patch merging layer concatenates the features of each group of 2×2 neighboring patches, and applies a linear layer on the $4C$ -dimensional concatenated features. This reduces the number of tokens by a multiple of $2 \times 2 = 4$. And the output dimension is set to $2C$. Swin Transformer blocks are applied afterwards for feature transformation, with the resolution kept at $H/8W/8$. This first block of patch merging and feature transformation is denoted as “Stage 2”. The procedure is repeated twice, as “Stage 3” and “Stage 4”, with output resolution of $H/16W/16$ and $H/32W/32$, respectively. These stages jointly produce a hierarchical representation.

Modifications: Traditional Swin Uses CNN-Based embedding. We replaced it with pure linear projection. Which helps us to convolutional inductive bias, making it more generalizable. Standard Swin Transformers Struggle with long-range dependencies. We introduce skip connections in MSA layers. Which improves feature propagation across windows.

Layer-by-Layer Breakdown of Custom Swin Transformer

Layer	Values/Operations	Purpose
Input (Segmented Image)	$(224 \times 224 \times 3)$	Disease-region- enhanced image

Layer	Values/Operations	Purpose
Patch Embedding (Linear Projection, No CNNs)	Converts $(224 \times 224 \times 3) \rightarrow (3136 \times 64)$	Transforms image into 64-dimensional tokens
Swin Transformer Block 1 (Custom W-MSA)	Self-attention in 7×7 windows	Captures local feature dependencies
Swin Transformer Block 2 (Custom SW-MSA)	Shifted windows attention (3×3 offset)	Ensures global information flow
Flatten	$(3136 \times 64) \rightarrow (128 \text{ neurons})$	Reduces feature map size
Fully Connected Layer	$128 \rightarrow \text{num_classes}$	Outputs final classification label

3.2.2 Design/Framework

The project's design includes a combination of individual model training and feature fusion. The U-net++ model was pre-trained and fine-tuned for this project. And the swin transformer trained independently with few modification. After training, their feature maps were extracted and concatenated to form a comprehensive feature vector for classification.

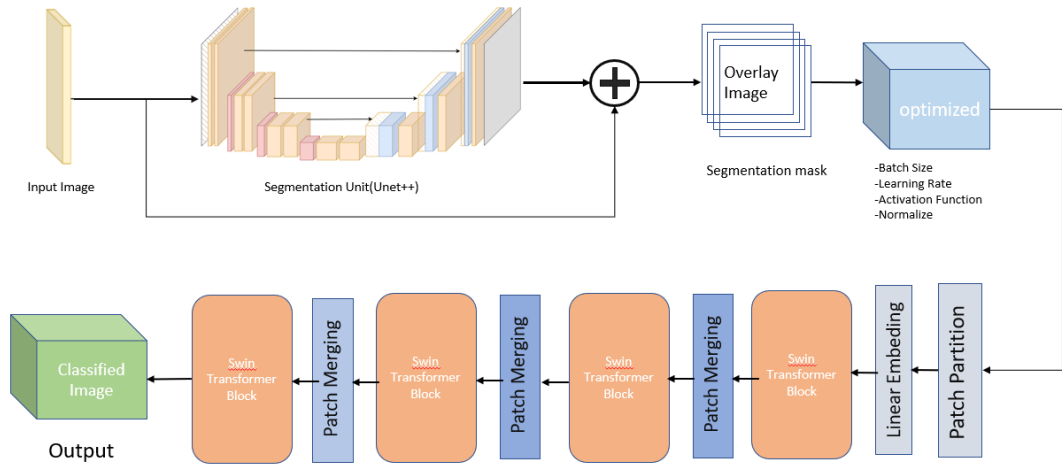


Figure 6: Full Methodology (The design of the Hybrid model featuring Unet++ and swin transformer)

3.2.3 Algorithm/Model Formulation

The feature fusion algorithm is designed as follows:

- Each model is trained independently.
 - Features from the segmentation unit layers are extracted and combined with the input image to create a mask for next input.
 - A fully connected layer is added after concatenation to form the final classification layer.
- The model is trained using the **Adam optimizer** with a learning rate of 0.001, batch size of 16, and **cross-entropy** loss function.

3.2.4 Experiment Setup

Training was carried out on a **train-validation-test split** of **70-15-15**.

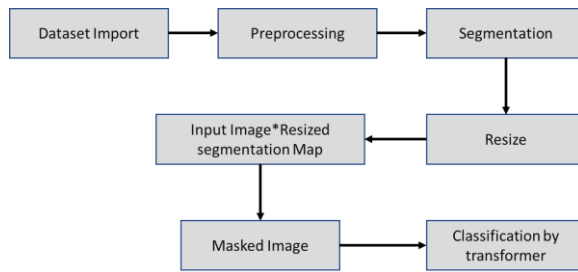


Figure 7: Experiment setup, including the dataset split, preprocessing steps, and model architecture.

3.3 Summary

In summary, the **Materials and Methods** outlined above explain the dataset acquisition, preprocessing, and model development processes. The proposed Hybrid model demonstrates the effectiveness of combining complementary architectures for OCT image classification tasks.

Chapter 4

Results and Discussion

4.1 Obtained Results

In this section, we present the results of our experiments with the proposed Hybrid model, and compare it against individual CNN architectures: MobileNetV3 and Vit-tiny. The results are summarized in terms of accuracy, precision, recall, and F1-score, showcasing the effectiveness of the proposed model.

Model	Accuracy (%)	Precision	Recall	F1-Score
MobileNetV3	94.39	0.92	0.92	0.94
Vit-tiny	95%	0.93	0.92	0.93
Custom Hybrid Model	96.72	0.9552	0.9497	0.9523

Table: Comparison against individual CNN architectures

4.2 In-Depth Result Analysis

Segmentation Maps: The probabilistic maps generated by the segmentation model normalized and colored using a heatmap. Combination of the original image and segmentation map allowing for a visual evaluation of alignment. These results provide a clear indication of how well the segmentation model identifies regions of interest.

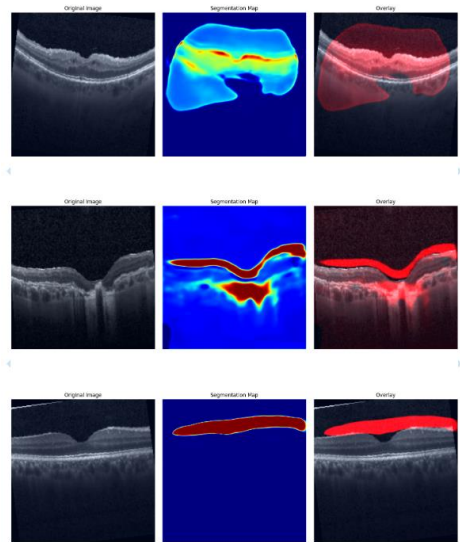


Fig 6: Segmentation Map Output

4.3

Classification Report and Confusion Matrix: The Classification report includes precision, recall and F1-score for each class in the test and validation datasets. Highlights class-level performance differences.

Classification Report (Test Data):				
	precision	recall	f1-score	support
CNV	0.98	0.98	0.98	5581
DME	0.97	0.97	0.97	1702
DRUSEN	0.91	0.87	0.89	1293
NORMAL	0.97	0.99	0.98	3947
accuracy			0.97	12523
macro avg	0.96	0.95	0.95	12523
weighted avg	0.97	0.97	0.97	12523

CNV and NORMAL classes exhibit the highest precision, recall, and F1-scores, reflecting strong model performance. DME has slightly lower recall compared to CNV and NORMAL, but still maintains good precision and F1-score. DRUSEN shows the lowest scores across precision and recall indicating that this class might be harder to classify accurately. Overall Accuracy 97%, demonstrating strong performance across the dataset. Macro Average highlights balanced performance across all classes and weighted average reflects overall performance considering the class distribution.

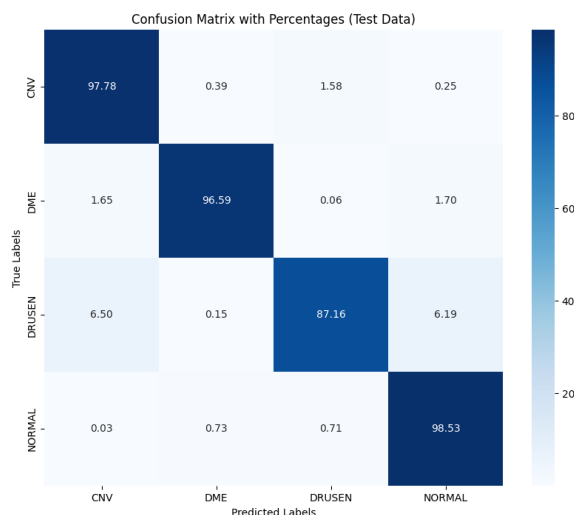


Fig 7 : Confusion Matrix of Test data

4.4

The confusion matrix visualizes the performance of the model by comparing the true class labels with the predicted class labels. Each cell represents the percentage of predictions for the corresponding pair of true and predicted labels. 97.78% of CNV samples are correctly classified as CNV. 96.59% of DME, 87.16% of DRUSEN and 98.53% for Normal are correctly classified.

4.5 Here are the report of classification and confusion Matrix for Validation Data:

Classification Report (Validation Data):				
	precision	recall	f1-score	support
CNV	0.98	0.98	0.98	5581
DME	0.97	0.96	0.96	1702
DRUSEN	0.92	0.87	0.89	1292
NORMAL	0.97	0.99	0.98	3948
accuracy			0.97	12523
macro avg	0.96	0.95	0.95	12523
weighted avg	0.97	0.97	0.97	12523

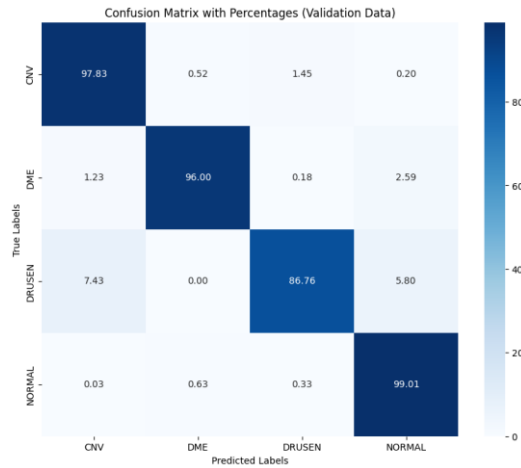


Fig 8: Confusion Matrix of Validation data

Performance Trends: Line plots for Loss and Accuracy over epochs. The plots illustrate the training and validation performance over epochs for loss and accuracy.

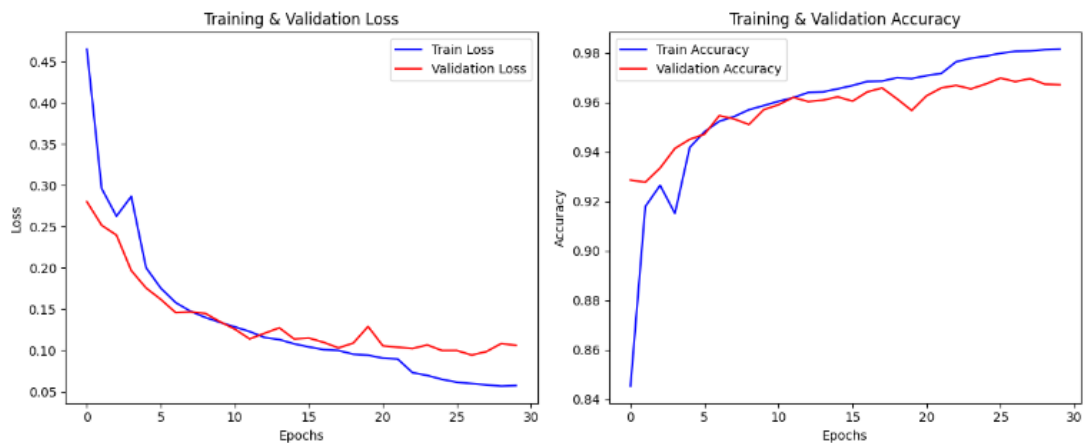


Fig 9: Performance Trends

Here the training loss(blue) and validation loss(red) are both decreasing over epochs. There is no significant divergence between the two, meaning the model is not overfitting. There are small fluctuations in validation loss, but that's normal due to batch variations. The training accuracy and validation accuracy are both increasing. Both are very close to each other, indicating good generalization and no major signs of overfitting.

ROC Curve: The plot represents the Receiver Operating Characteristic(ROC) curve for a multi-class classification problem. Each class is evaluated independently using a one-vs-rest approach.

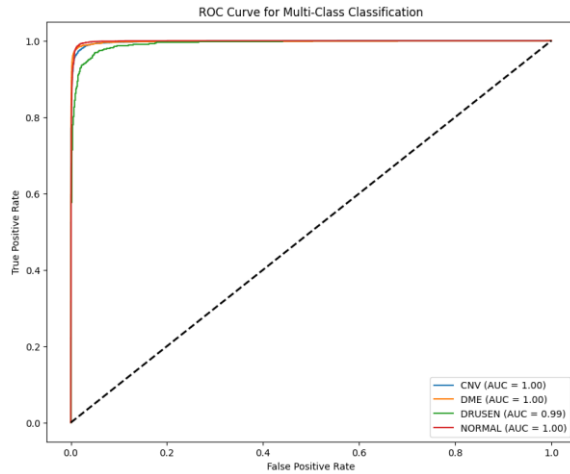


Fig 10: ROC Curve

Each colored line corresponds to one of the four classes (CNV, DME, DRUSEN, NORMAL). A perfect classifier will have a curve that hugs the top-left corner, achieving 1.0 TPR with minimal FPR. CNV: AUC=1.00, DME: AUC = 0.99, DRUSEN: AUC =0.99, NORMAL: AUC = 1.00. Which are indicating excellent performance in distinguishing each class from the others. Overall model performance is:

Accuracy: 0.9672
Precision: 0.9552
Recall: 0.9497
F1-Score: 0.9523

4.4 Discussion

This section provides a synthesis of the findings from the obtained results and the analysis. Key points include:

- The significant advantages of the Hybrid approach in enhancing model accuracy and robustness.
- Challenges faced by individual models in varying different image conditions, such as resolution clinical image complexity.

- The implications of the findings for efficient eye care applications and future research directions.

Reference	Method	Accuracy	Comments
Paraschive Elena-Anca (2023) [1]	DenseNet21	95.56%	Enhances feature propagation and reuse through dense connections, improving efficiency with fewer parameters.
	InceptionResNet	93.18%	Combines the Inception module's feature extraction with ResNet's shortcut connections for optimized performance
	Covolutional Layers-Based model	93%	
Aya Adel, Mona M.Soliman, Nour Eldeen M. Khalifa, and Khaled Mostafa.(2020) [9]	Inception V3 based model	93%	Employs factorized convolutions and auxiliary classifiers to improve accuracy while reducing computational cost.

Table: Comparison with State-of-the-Art Research

4.5 Summary

In summary, the experiments conducted demonstrate the effectiveness of the Hybrid model in achieving high accuracy in Retinal image classification tasks while highlighting the importance of model selection and feature fusion. The results indicate that leveraging multiple architectures can lead to more robust and accurate systems, paving the way for advancements in medical image technologies.

Chapter 5

Conclusion

5.1 Overall Contributions

Nowadays, people are exposed to screens for long hours. This leads to a rise in retinal diseases. Therefore, it is necessary to detect retinal diseases early. A practical CAD(computer-aided diagnostic) system is required to help ophthalmology.

This study has made significant contributions to the field of retinal disease recognition by proposing a novel feature fusion model, that combines the strengths of two advanced deep learning architectures. The key contributions are as follows:

- **Innovative Model Development:** Our hybrid model showcases the effectiveness of feature fusion in enhancing retinal disease classification accuracy, achieving an impressive accuracy of 97%. This performance surpasses that of individual models, demonstrating the potential of integrating multiple architectures to leverage their complementary strengths.
- **Comprehensive Performance Evaluation:** A thorough analysis of various models was conducted, including assessments based on accuracy, precision, recall, F1-score, and computational efficiency. This comprehensive evaluation provides valuable insights into the operational capabilities of each model and highlights the advantages of the proposed feature fusion approach.
- **Practical Applications and Interface Development:** The successful implementation of a real-time retinal disease recognition system using provides a practical application for the findings. This system is clinical friendly and eliminates the experts help for quick diagnosis.

5.2 Limitations and Future Works

Despite the successes achieved, this research also acknowledges certain limitations and outlines potential avenues for future work:

Model Complexity and Resource Requirements: While **Our hybrid model** offers high accuracy, its computational demands may restrict its deployment in resource-constrained environments. Future research could focus on optimizing the model through techniques such as model pruning and quantization to enhance efficiency without sacrificing performance.

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Appendix A

Mapping of Course and Program Outcomes

ICE496

Course Objective and Outcomes:

Course Objective: The Capstone project serves as a culminating experience, enabling students to demonstrate proficiency across multiple learning outcomes established by their degree program. This project integrates various disciplines, emphasizing the application of professional skills in real-world scenarios.

Course Outcomes (COs): Upon completion of this course, students will be able to:

- **CO1:** Identify and examine various problem domains through a literature review. In this project, we conducted extensive research on retinal disease classification technologies, justifying and finalizing our problem statement: improving the accuracy and robustness through a hybrid model. We developed a comprehensive project proposal outlining our approach and methodologies.
- **CO2:** Perform and complete requirement analysis. For our finalized project, we outlined design methodologies, focusing on the integration of multiple architectures to enhance model performance.
- **CO3:** Design hardware and/or software for the finalized project, incorporating societal, environmental, and ethical considerations. We built a retinal disease diagnosis system utilizing pre-trained models to minimize resource consumption while ensuring high accuracy. The project management principles were applied to manage the development process effectively.
- **CO4:** Utilize various visualization tools to write and present a technical report. Throughout the project, we employed visualization techniques to illustrate our findings, including performance metrics and model architecture. Our final presentation incorporated evaluation metrics, showcasing the ethical, societal, and economic implications of our gesture recognition system.

CO	Details	Knowledge Profile (K)	Engineering problem (EP)
CO1	The project began with the identification and examination of various problem domains within the field of medical image processing . We conducted a literature review to justify and finalize the problem statement, focusing on the need for improved accuracy and efficiency in retinal disease detection. This led to the development of a project proposal outlining our objectives and methodologies.	(i) Identify a real-life problem [K1, K2, K3, K4] K1: Understanding the theoretical underpinnings and deep learning techniques. K2: Applying mathematical and statistical analysis to evaluate the methods. K3: Grasping the fundamental engineering principles that guide the design of deep learning models. K4: Utilizing advanced knowledge to address current limitations in this technologies.	(i) Identify a real-life problem [EP1, EP2, EP3, EP4, EP5, EP6, EP7] EP1: Depth of knowledge required: Understanding the complexities of OCT systems and their limitations. EP2: Range of conflicting requirements: Balancing accuracy, speed, and resource consumption in model design. EP3: Depth of analysis required: Assessing various models and techniques to select the most appropriate for the project. EP5: Extent of applicable codes: Ensuring adherence to ethical standards and best practices in AI. EP6: Stakeholder involvement: Engaging with team members and mentors to refine the project proposal. EP7: Interdependence: Collaborating with team

			members to consolidate ideas and finalize the problem statement.
CO2	The design methodology involved a systematic approach to addressing challenges, integrating deep learning and computer vision principles. Key steps included model selection, training strategy, feature extraction, feature fusion, and final classification. we enhanced robustness and accuracy under varied conditions.	(i) Design methodology for complex systems [K5, K6] K5: Design-based engineering principles for systems integration and development. K6: Research literature supporting model selection and fusion techniques.	(i) Defining the design and methodology [EP1, EP2, EP3, EP4, EP5, EP6, EP7] EP1: Depth of knowledge required: Understanding of CNN architectures and their applications. EP2: Range of conflicting requirements: Balancing accuracy, computational efficiency, and real-time processing needs. EP3: Depth of analysis required: Analyzing model performance metrics and adjusting training strategies. EP5: Extent of applicable codes: Adhering to ethical considerations in AI and machine learning. EP6: Stakeholder involvement: Ensuring system usability and effectiveness for end-users. EP7: Interdependence:

			Integration of multiple models and their outputs to enhance overall performance.
CO3	The project involved designing and building the retinal disease classification system, integrating multiple architectures to create the hybrid model. This system incorporates both hardware (High resolution camera) and software components (deep learning frameworks) to identify the disease accurately. The design process accounted for societal and ethical considerations, ensuring the system is user-friendly and accessible.	(i) Designing and building systems [K7, K8] K7: Engineering practices for system integration and development. K8: Ethical considerations in technology deployment and use.	(i) Implementing the deliverable system [EP1, EP2, EP3, EP4, EP5, EP6, EP7] EP1: Depth of knowledge required: Understanding the integration of hardware and software systems. EP2: Range of conflicting requirements: Ensuring the system meets performance standards while being user-friendly. EP3: Depth of analysis required: Analyzing the performance of the integrated system in real-time scenarios. EP5: Extent of applicable codes: Compliance with industry standards for software and hardware integration. EP6: Stakeholder

			involvement: Engaging with users to gather feedback for iterative improvements. EP7: Interdependence: Understanding the relationship between hardware performance and software algorithms in achieving accuracy.
CO4	The project required the use of different visualization tools to present technical reports and results effectively. We created a comprehensive technical report detailing the methodologies, results, and implications of the system. Additionally, we prepared presentations that incorporated various evaluation metrics, including efficiency, cost, and societal impacts.	(i) Utilizing visualization tools and preparing reports [K9] K9: Proficiency in using data visualization tools and techniques to convey complex information.	(i) Presenting findings and evaluations [EP1, EP2, EP3, EP4, EP5, EP6, EP7] EP1: Depth of knowledge required: Understanding effective communication strategies for technical information. EP2: Range of conflicting requirements: Balancing technical detail with clarity for diverse audiences. EP3: Depth of analysis required: Analyzing data and results to identify key insights and implications. EP4: Familiarity of issues: Addressing potential misunderstandings that may arise from technical jargon. EP5: Extent of applicable

			codes: Adhering to guidelines for technical reporting and presentations. EP6: Stakeholder involvement: Engaging with peers and faculty to refine presentations based on feedback. EP7: Interdependence: Coordinating between technical writing and visual representation for effective communication.
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The End